

Morris Lab Abaqus Hoffman2 Cluster SOP

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1 All Required Software and Tutorials

Before you begin, make sure you have the following (keep in mind this is all Windows interface):

- A computer/laptop - duh.
- A Hoffman2 account - make sure you have access to your lab's login.
- PuTTY - the SSH client to access Hoffman2. See the download link [here](#).
- VcXsrv - to render the software you need through Hoffman2 Clusters. Access the link to download [here](#).
- Cisco VPN - to access UCLA's VPN. See [here](#) for the access link.

Here's some quick access guides to certain websites that you may find helpful:

- Hoffman2 How-tos: The official documentation and the GitHub Guide by Chris German for Biostatisticians, but still applicable nonetheless.
- Abaqus How-tos: YouTube Tutorials

2 Setting Up VcXsrv

VcXsrv by itself cannot handle the graphics of Abaqus. So, you'll need to do some bypassing and playing around. Open XLaunch to set up the system. Then, do the following steps:

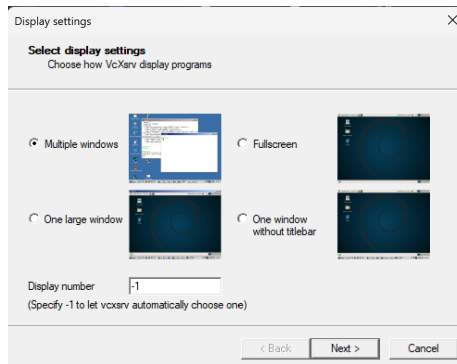


Figure 1: Click multiple windows and set display number to -1 to let VcXsrv decide.

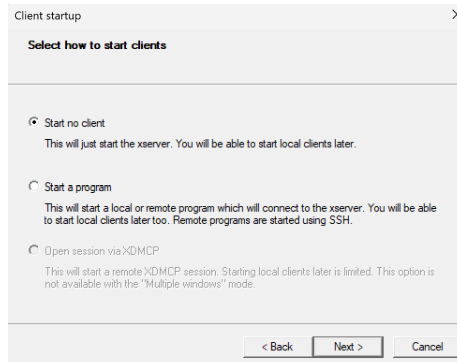


Figure 2: Select start no client.

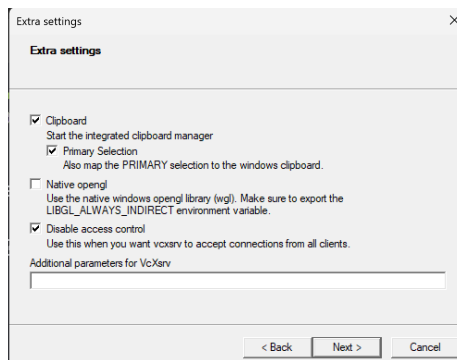


Figure 3: Select Clipboard and Disable Access Control. Do not select Native opengl

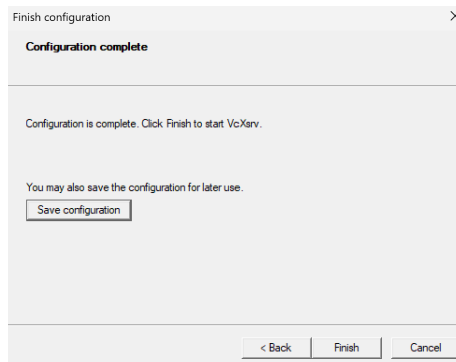


Figure 4: Save configuration and click Finish

Congratulations! VcXsrv is ready for use!

3 Accessing Abaqus

First, connect to Cisco first. Then, check if VcXsrv is running by seeing if XLaunch shows up when you click the ‘Show Hidden Icons’ button, and it shows up like this:

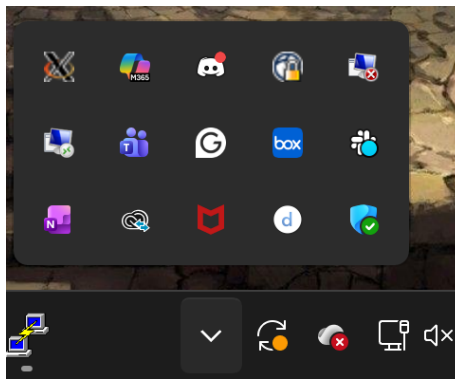


Figure 5: How it should look when XLaunch is open

Next, open PuTTY and type in your username. Then go to the left navigation menu, find SSH, expand it, look for X11 and click on it. Then, check the box that says ‘Enable X11 forwarding’ and click open.

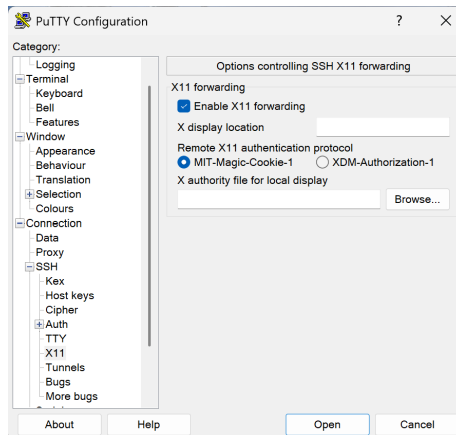


Figure 6: PuTTY X11 forwarding steps

Once you're in, send in a session request by typing:

```
qrsh -l h\_data=16G,h\_rt:01:00:00
```

Now that you're in a computing node, you'll type in the following three lines:

```
export LIBGL_ALWAYS_INDIRECT=1
module load abaqus
abaqus cae
```

Once this is all done, Abaqus should open up like so:

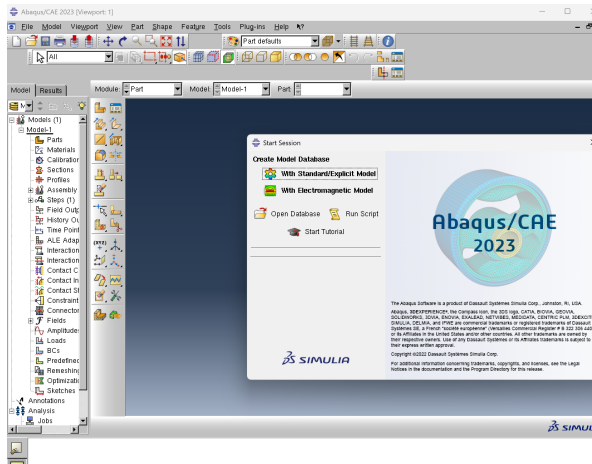


Figure 7: Abaqus interface using VcXsrv

All there's left to do is to upload your files as needed onto your remote desktop, and start using Abaqus. Happy modeling!